

Volume Comparisons in the Cerebellar Complex of Primates

II. Cerebellar Nuclei

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Abstract. Volumes of medial, interposed, and lateral cerebellar nuclei (MCN, ICN, and LCN) were measured in Insectivora, Scandentia, and Primates, including man. The relative size of the nuclei was expressed in size indices. Insectivora had by far the smallest cerebellar nuclei. The simians, in general, had larger cerebellar nuclei than the prosimians, but there was considerable overlap. From Insectivora to man, the MCN was the least progressive and the LCN the most progressive. The indices are expected to reflect the relative size of the three longitudinal zones of the cerebellum (vermis/MCN, pars intermedialis/ICN, hemisphere/LCN). They, together with those of the ventral pons and cerebellum (part I), are discussed in relation to the predominant locomotor pattern of a species, and with reference to evolutionary trends in primate phylogeny.

Introduction

The cerebellar nuclei are embedded in the deep white matter of the cerebellum as three masses of grey matter. These are, going laterally, the medial (MCN), the interposed (ICN), and the lateral (dentate) nucleus (LCN). The efferent fiber bundles, conveying cerebellar information to the centers of motor coordination, arise from these nuclei. Since the study on the cerebellar cortico-nuclear projections by Jansen and Brodal [1940], it is generally accepted that the cere-

bellum has three principal components (or three longitudinal zones): the vermis-medialis component, the pars intermedialis-interpositus component, and the hemisphere-lateralis component [Kuhlenbeck, 1975; Eccles, 1982]. The various locomotory patterns in primates, ranging from simple automatic movements to very complex voluntary movements, are made possible by close interaction of these three zones. The lateral zone is involved in the enormously complex organization and control of voluntary movement [Eccles, 1982], and seems to be the most pro-

gressive in primate phylogeny. The percentage of the cerebellar hemispheres relative to the total cerebellum increases from about 34% in Insectivora [unpubl. data] to about 89% in man (see part I).

In this paper, the size of the cerebellar nuclei will be considered an indication of the variations in the three longitudinal zones among primates, and will be compared from phylogenetic and ecoethological points of view. The results will be compared with those on the ventral pons (VPo) and the entire cerebellum given in part I [Matano et al., 1985].

In the cat [Flood and Jansen, 1961] and in *Macaca mulatta* [Courville and Cooper, 1970], the ICN is divided into anterior and posterior parts. Carpenter [1976] suggested that the anterior part of the ICN is homologous to the emboliformis nucleus in man, while the posterior part is homologous to the globosus nucleus. In the present study, the ICN was treated as a single nucleus in all the species investigated, including man.

Materials and Methods

Volumes of the cerebellar nuclei were measured in the brains of 90 individuals of 56 species belonging to Insectivora (10 species), Scandentia (2 species), and Primates (18 species of prosimians and 26 species of simians). Details are listed in table I. More detailed information on the methods (1) of fixation and preparation of the brains and sections, (2) of the determination of volumes and their conversion to fresh values, and (3) of interspecific and interstructural comparisons, are given in part I, and by Stephan et al. [1981, 1985].

Four methods were used for the size comparisons: (1) determination of the percentage of the cerebellar nuclei as a whole relative to the cerebellum as a whole (table II, column 1); (2) percentage of each nucleus relative to the cerebellar nuclei as a whole (table II, columns 2–4); (3) interstructural regression analyses of

the cerebellar nuclei with other structures in double logarithmic plots (table IV); and (4) size indices, which evaluate size differences in isoponderous species (table II, columns 5–8). In the fourth method, the cerebellar nuclei volumes were plotted against body weights in double logarithmic scales, and least square regression lines were determined within narrowly related groups (subfamilies, families). The slopes converged at a value similar to that of the whole cerebellum and VPo (0.67) and, therefore, this slope was assigned to the reference lines. These lines were drawn through the average of the prosimian values ($n = 18$). Their equations are:

$\log \text{TCN} = -0.423 + 0.67 \log \text{BoW}$ for the total cerebellar nuclei,

$\log \text{MCN} = -0.928 + 0.67 \log \text{BoW}$ for the medial nucleus,

$\log \text{ICN} = -0.971 + 0.67 \log \text{BoW}$ for the interposed nucleus,

$\log \text{LCN} = -0.815 + 0.67 \log \text{BoW}$ for the lateral nucleus.

The reference line for the LCN is shown in figure 1. Any value on the reference lines is considered to represent the average cerebellar nucleus volume for a typical prosimian and has an index of 1. The distances from the reference line are expressed by deviations from 1 and are a measure of the degree of deviation of the cerebellar nuclei volumes from those of typical prosimians. The indices are listed in table II, columns 5–8. Average values for taxonomic groups and locomotor types (indices and percentages) are given in table III.

Results

For *interspecific* comparisons, the size index method (regressing log cerebellar nuclei vs. log body weight) is more appropriate, whereas for *interstructural* comparisons, percentages and interstructural regressions may give more informative results.

Size Indices

The indices based on the prosimian average (= 1) are given in table II, and scaled in figures 2–8; means are given in table III.

Table I. Body weights (BoW); volumes of total cerebellar nuclei (TCN), of medial (MCN), interpositus (ICN), and lateral (LCN) cerebellar nuclei; and ecoethological characters

	Specimens	BoW g	Cerebellar nuclei, mm ³				Ecoethological characteristics			
	n		TCN	MCN	ICN	LCN				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
Insectivora										
0577	<i>Tenrec ecaudatus</i>	1	832	9.82	2.06	4.43	3.33	N	1	TQ
0579	<i>Setifer setosus</i>	1	248	4.19	1.20	1.78	1.21	N	1	TQ
0581	<i>Hemicentetes semispin.</i>	1	110	3.23	0.55	1.52	1.16	N	1	TQ
0585	<i>Echinops telfairi</i>	1	87.5	1.71	0.39	0.87	0.46	N	1	TQ
0695	<i>Erinaceus europaeus</i>	1	860	12.65	3.30	5.53	3.83	N	1	TQ
0715	<i>Sorex araneus</i>	1	10.2	0.58	0.18	0.18	0.21	N	1	TQ
0769	<i>Sorex minutus</i>	1	4.42	0.35	0.09	0.13	0.13	N	1	TQ
0943	<i>Crocidura flavescens</i>	1	33.1	1.17	0.27	0.42	0.47	N	1	TQ
1070	<i>Crocidura r. russula</i>	1	11.1	0.58	0.12	0.23	0.23	N	1	TQ
1137	<i>Suncus murinus</i>	1	33.0	1.06	0.23	0.38	0.44	N	1	TQ
Scandentia										
1263	<i>Tupaia glis</i>	2	170	12.46	3.93	3.40	5.13	D	1	SAQ
1289	<i>Urogale everetti</i>	1	275	15.07	4.13	3.94	7.00	D	1	SAQ
Prosimians										
3199	<i>Microcebus murinus</i>	2	54	4.74	1.49	1.14	2.11	N	1	AQP
3203	<i>Cheirogaleus major</i>	2	450	19.45	5.65	3.85	9.95	N	2	AQP
3205	<i>Cheirogaleus medius</i>	1	177	9.45	3.24	1.94	4.27	N	2	AQP
3215	<i>Lemur fulvus</i>	2	1,400	86.28	26.65	27.58	32.05	D	3	AQP
3227	<i>Varecia variegata</i>	1	3,000	94.67	28.39	28.83	37.45	D	3	AQP
3239	<i>Lepilemur ruficaudatus</i>	1	915	23.07	7.31	6.41	9.35	N	3	VCL
3243	<i>Avahi l. laniger</i>	1	1,285	37.89	12.15	11.28	14.46	N	3	VCL
3244	<i>Avahi l. occidentalis</i>	1	860	29.66	8.18	10.18	11.30	N	3	VCL
3247	<i>Propithecus verreauxi</i>	2	3,480	97.12	26.72	30.94	39.46	D	3	VCL
3249	<i>Indri indri</i>	2	6,250	141.30	38.03	46.96	56.31	D	3	VCL
3251	<i>Daubentonia madagasc.</i>	1	2,800	126.08	28.94	36.11	61.03	N	1	AQP
3253	<i>Loris tardigradus</i>	2	322	17.99	6.65	5.58	5.76	N	1	SCQ
3255	<i>Nycticebus coucang</i>	2	800	33.75	10.96	8.76	14.03	N	1	SCQ
3259	<i>Perodicticus potto</i>	2	1,150	29.44	8.76	8.43	12.25	N	2	SCQ
3265	<i>Galago senegalensis</i>	2	186	12.60	4.43	3.77	4.40	N	2	VCL
3267	<i>Otolemur crassicaudat.</i>	2	850	25.90	9.63	6.63	9.64	N	2	VCL
3275	<i>Galagoides demidoff</i>	2	81	8.22	3.08	2.08	3.06	N	1	VCL
3282	<i>Tarsius spp.</i>	4	125	8.06	2.87	1.82	3.37	N	1	VCL
Simians										
3287	<i>Callithrix jacchus</i>	2	280	20.96	7.59	5.44	7.93	D	2	NAQ
3289	<i>Cebuella pygmaea</i>	2	120	11.48	3.93	3.20	4.35	D	2	NAQ
3305	<i>Saguinus midas</i>	2	340	21.57	6.88	6.94	7.75	D	2	NAQ
3311	<i>Saguinus oedipus</i>	2	380	24.60	8.61	7.19	8.80	D	2	NAQ

Table I (continued)

	Speci- mens n	BoW g	Cerebellar nuclei, mm ³				Ecoethological characteristics		
			TCN	MCN	ICN	LCN	(7)	(8)	(9)
3317 <i>Cebus albifrons</i>	1	3,100	173.18	34.68	38.98	99.52	D	2	NAQ
3325 <i>Aotus trivirgatus</i>	2	830	41.45	13.01	11.42	17.02	N	2	NAQ
3327 <i>Callicebus moloch</i>	2	900	61.41	20.70	19.47	21.24	D	2	NAQ
3335 <i>Saimiri sciureus</i>	2	660	73.22	24.09	21.93	27.20	D	2	NAQ
3341 <i>Pithecia monachus</i>	2	1,500	109.36	23.91	31.78	53.67	D	2	NAQ
3363 <i>Alouatta seniculus</i>	2	6,400	113.79	27.47	26.57	59.75	D	3	NSB
3371 <i>Ateles geoffroyi</i>	1	8,000	209.17	34.63	59.75	114.79	D	2	NSB
3379 <i>Lagothrix lagothricha</i>	2	5,200	212.95	44.15	60.23	108.57	D	2	NSB
3391 <i>Macaca mulatta</i>	2	7,800	235.31	53.24	75.13	106.94	D	2	OTQ
3407 <i>Cercocebus albigena</i>	2	7,900	232.63	44.70	74.60	113.33	D	2	OAQ
3415 <i>Papio anubis</i>	2	25,000	379.04	58.78	99.91	220.35	D	2	OTQ
3433 <i>Cercopithecus ascanius</i>	2	3,400	143.98	34.61	41.70	67.67	D	2	OAQ
3453 <i>Cercopithecus mitis</i>	1	6,300	171.16	34.23	54.55	82.38	D	2	OAQ
3469 <i>Miopithecus talapoin</i>	2	1,200	78.95	18.69	26.42	33.84	D	2	OAQ
3473 <i>Erythrocebus patas</i>	2	7,800	295.09	73.71	95.07	126.31	D	2	OTQ
3477 <i>Colobus badius</i>	2	7,000	190.92	34.64	57.76	98.52	D	3	OSB
3493 <i>Pygathrix nemaeus</i>	1	7,500	265.75	47.94	80.76	137.05	D	3	OSB
3499 <i>Nasalis larvatus</i>	1	14,000	340.36	85.55	105.48	149.33	D	3	OSB
3539 <i>Hylobates lar</i>	1	5,700	242.09	35.41	65.60	141.08	D	2	OTB
3549 <i>Pan troglodytes</i>	1	46,000	601.42	58.32	87.27	455.83	D	2	OMB
3551 <i>Gorilla gorilla</i>	2	105,000	740.40	93.44	112.36	534.60	D	3	OMB
3553 <i>Homo sapiens</i>	2	65,000	1,513.78	156.40	190.48	1,166.90	D		SBP

(7) D = Diurnal; N = nocturnal.

(8) 1 = Insectivores; 2 = frugivores; 3 = folivores.

(9) AQP = Arboreal quadrupedal prosimians; NAQ = New World arboreal quadrupedal simians; NSB = New World semibrachiators; OAQ = Old World arboreal quadrupedal simians; OMB = Old World modified brachiators; OSB = Old World semibrachiators; OTB = Old World true brachiators; OTQ = Old World terrestrial quadrupedal simians; SAQ = semiarboreal quadrupeds; SBP = specialized bipedal hominid; SCQ = slow-climbing quadrupedal prosimians; TQ = terrestrial quadrupeds; VCL = vertical clinging and leaping prosimians.

Total Cerebellar Nuclei. The average TCN indices were 0.31 for Insectivora, 0.99 for Scandentia, 1.52 for non-human simians, and 2.39 for man. The lowest index (0.23) was found in *Echinops*, i.e., one of the 'basal' Insectivora. This index indicates cerebellar

nuclei of about 1/4 that of an average prosimian. Scandentia fully overlap with prosimians and have a similar average. Simians partly overlap with prosimians. In prosimians, the lowest index was found in *Lepilemur*, the highest in Lemurinae and in *Daubentonina*. In

Table II. Percentages and indices of total cerebellar nuclei (TCN), and of medial (MCN), interpositus (ICN), and lateral (LCN) cerebellar nuclei (CER = total cerebellum)

		Percentages				Indices			
		TCN CER	MCN TCN	ICN TCN	LCN TCN	TCN	MCN	ICN	LCN
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Insectivora¹									
0577	<i>Tenrec ecaudatus</i>	3.18	21.0	45.1	33.9	0.29	0.19	0.46	0.24
0579	<i>Setifer setosus</i>	2.95	28.6	42.6	28.8	0.28	0.25	0.42	0.20
0581	<i>Hemicentetes semispin.</i>	2.76	16.9	47.2	35.9	0.37	0.20	0.61	0.32
0585	<i>Echinops telfairi</i>	2.96	22.7	50.6	26.7	0.23	0.16	0.40	0.15
0695	<i>Erinaceus europaeus</i>	3.21	26.1	43.7	30.2	0.36	0.30	0.56	0.27
0715	<i>Sorex araneus</i>	2.63	30.9	31.9	37.2	0.32	0.32	0.36	0.29
0769	<i>Sorex minutus</i>	2.96	26.6	35.6	37.8	0.35	0.29	0.44	0.32
0943	<i>Crocidura flavescens</i>	2.21	23.4	36.3	40.3	0.30	0.22	0.38	0.30
1070	<i>Crocidura r. russula</i>	2.69	20.2	39.9	39.9	0.30	0.20	0.43	0.30
1137	<i>Suncus murinus</i>	2.52	22.1	36.1	41.8	0.27	0.19	0.34	0.28
Scandentia									
1263	<i>Tupaia glis</i>	3.44	31.5	27.3	41.2	1.06	1.07	1.02	1.07
1289	<i>Urogale everetti</i>	3.05	27.4	26.1	46.5	0.93	0.81	0.86	1.06
Prosimians									
3199	<i>Microcebus murinus</i>	2.08	31.4	24.1	44.5	0.87	0.87	0.74	0.95
3203	<i>Cheirogaleus major</i>	2.12	29.0	19.8	51.2	0.86	0.80	0.60	1.08
3205	<i>Cheirogaleus medius</i>	2.23	34.3	20.5	45.2	0.78	0.86	0.57	0.87
3215	<i>Lemur fulvus</i>	2.70	30.9	32.0	37.1	1.78	1.76	2.01	1.63
3227	<i>Varecia variegata</i>	2.30	30.0	30.4	39.6	1.17	1.13	1.26	1.14
3239	<i>Lepilemur ruficaudatus</i>	2.03	31.7	27.8	40.5	0.63	0.64	0.62	0.63
3243	<i>Avahi l. laniger</i>	2.65	32.1	29.8	38.1	0.83	0.85	0.87	0.78
3244	<i>Avahi l. occidentalis</i>	2.24	27.6	34.3	38.1	0.85	0.75	1.03	0.80
3247	<i>Propithecus verreauxi</i>	2.56	27.5	31.9	40.6	1.09	0.96	1.23	1.09
3249	<i>Indri indri</i>	2.66	26.9	33.2	39.9	1.07	0.92	1.26	1.05
3251	<i>Daubentonia madagasc.</i>	2.05	23.0	28.6	48.4	1.64	1.20	1.66	1.95
3253	<i>Loris tardigradus</i>	2.57	37.0	31.0	32.0	0.99	1.18	1.09	0.79
3255	<i>Nycticebus coucang</i>	2.71	32.5	25.9	41.6	1.01	1.05	0.93	1.04
3259	<i>Perodicticus potto</i>	1.82	29.8	28.6	41.6	0.69	0.66	0.70	0.71
3265	<i>Galago senegalensis</i>	1.93	35.2	29.9	34.9	1.01	1.13	1.06	0.87
3267	<i>Otolemur crassicaudat.</i>	1.90	37.2	25.6	37.2	0.75	0.89	0.68	0.69
3275	<i>Galagoides demidoff</i>	2.06	37.5	25.3	37.2	1.14	1.38	1.02	1.05
3282	<i>Tarsius spp.</i>	1.96	35.6	22.6	41.8	0.84	0.96	0.67	0.87
Simians									
3287	<i>Callithrix jacchus</i>	2.92	36.2	26.0	37.8	1.27	1.48	1.17	1.19
3289	<i>Cebuella pygmaea</i>	2.77	34.2	27.9	37.9	1.23	1.35	1.21	1.15
3305	<i>Saguinus midas</i>	2.12	31.9	32.2	35.9	1.15	1.17	1.31	1.02

Table II (continued)

		Percentages				Indices			
		$\frac{\text{TCN}}{\text{CER}}$	$\frac{\text{MCN}}{\text{TCN}}$	$\frac{\text{ICN}}{\text{TCN}}$	$\frac{\text{LCN}}{\text{TCN}}$	TCN	MCN	ICN	LCN
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
3311	<i>Saguinus oedipus</i>	2.64	35.0	29.2	35.8	1.22	1.36	1.26	1.07
3317	<i>Cebus albifrons</i>	2.41	20.0	22.5	57.5	2.10	1.35	1.67	2.98
3325	<i>Aotus trivirgatus</i>	2.35	31.4	27.5	41.1	1.22	1.22	1.18	1.23
3327	<i>Callicebus moloch</i>	3.95	33.7	31.7	34.6	1.71	1.84	1.91	1.45
3335	<i>Saimiri sciureus</i>	3.45	32.9	29.9	37.2	2.50	2.64	2.65	2.29
3341	<i>Pithecia monachus</i>	2.96	21.9	29.0	49.1	2.16	1.51	2.21	2.61
3363	<i>Alouatta seniculus</i>	2.15	24.1	23.4	52.5	0.85	0.66	0.70	1.10
3371	<i>Ateles geoffroyi</i>	1.80	16.5	28.6	54.9	1.34	0.71	1.36	1.82
3379	<i>Lagothrix lagothricha</i>	2.02	20.7	28.3	51.0	1.83	1.21	1.82	2.30
3391	<i>Macaca mulatta</i>	2.83	22.6	31.9	45.5	1.54	1.11	1.73	1.72
3407	<i>Cercocebus albigena</i>	2.36	19.2	32.1	48.7	1.51	0.93	1.71	1.81
3415	<i>Papio anubis</i>	2.30	15.5	26.4	58.1	1.14	0.56	1.06	1.63
3433	<i>Cercopithecus ascanius</i>	2.71	24.0	29.0	47.0	1.64	1.26	1.68	1.90
3453	<i>Cercopithecus mitis</i>	2.74	20.0	31.9	48.1	1.29	0.83	1.45	1.53
3469	<i>Miopithecus talapoin</i>	2.55	23.7	33.5	42.8	1.81	1.37	2.14	1.91
3473	<i>Erythrocebus patas</i>	3.72	25.0	32.2	42.8	1.93	1.54	2.20	2.04
3477	<i>Colobus badius</i>	2.39	18.1	30.3	51.6	1.34	0.78	1.43	1.71
3493	<i>Pygathrix nemaeus</i>	3.61	18.0	30.4	51.6	1.78	1.03	1.91	2.27
3499	<i>Nasalis larvatus</i>	3.05	25.1	31.0	43.9	1.50	1.21	1.65	1.63
3539	<i>Hylobates lar</i>	2.22	14.6	27.1	58.3	1.95	0.91	1.87	2.81
3549	<i>Pan troglodytes</i>	1.52	9.7	14.5	75.8	1.20	0.37	0.61	2.24
3551	<i>Gorilla gorilla</i>	1.16	12.6	15.2	72.2	0.85	0.34	0.45	1.51
3553	<i>Homo sapiens</i>	1.19	10.3	12.6	77.1	2.39	0.79	1.06	4.54

¹ In Insectivora the ventral pons is included in the cerebellum.

simians, the lowest values were found in *Alouatta* and *Gorilla*, the highest in *Saimiri*, and the second highest in man. There is a general trend for relative size of the TCN to increase from Insectivora through prosimians and non-human simians up to man. In man, the TCN are about 10 times larger than in *Echinops* of (theoretically) equal body weight. The various components (MCN, ICN, LCN) behave differently (fig. 2–4).

Medial Cerebellar Nucleus (fig. 2, 4). The average MCN indices were 0.23 for Insectivora, 0.94 for Scandentia, 1.15 for non-human simians, and 0.79 for man. Scandentia fully overlap with primates, and in primates, simians fully overlap with prosimians; there is nearly no change in the average values. The distinctly lowest indices again were found in Insectivora. In prosimians, Lemurinae, *Galagoides*, and *Daubentonia* have

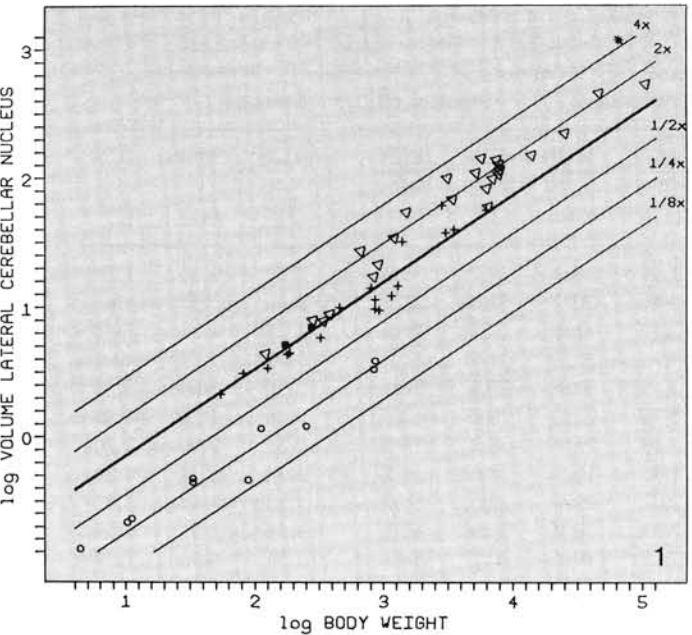
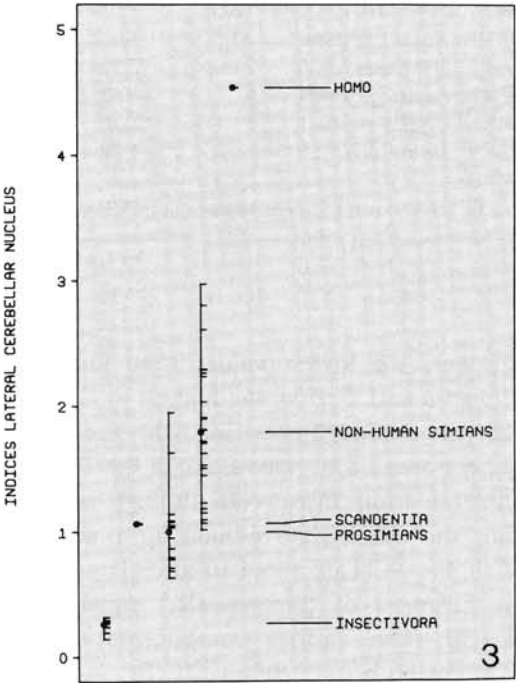
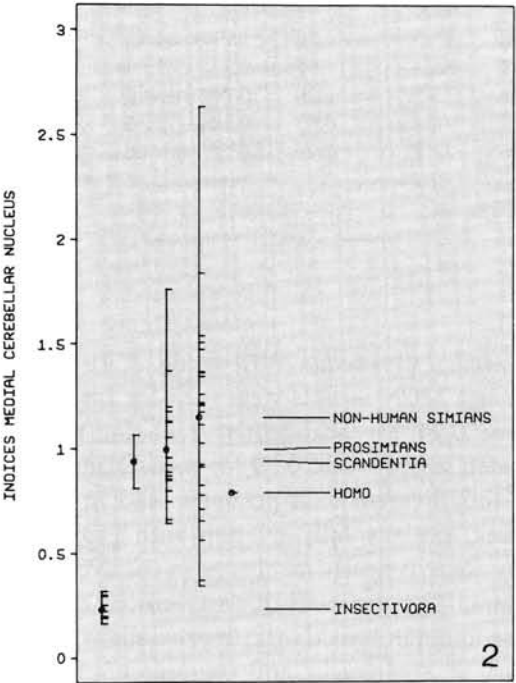


Fig. 1. Lateral cerebellar nucleus (LCN) volume plotted against body weight (BoW) on a double logarithmic scale. The volumes are in mm³; the weights in grams. The thick reference line is placed through the average for prosimians. Its slope is determined by least square regression analyses within primate families and subfamilies. The thinner parallel lines are in distances of 0.301 (i.e. in multiples and fractions of 2) and correspond to the indices given in table II, and scaled in figures 3 and 7. ○ = Insectivora; ■ = Scandentia; + = prosimians; ▽ = non-human simians; ★ = man.



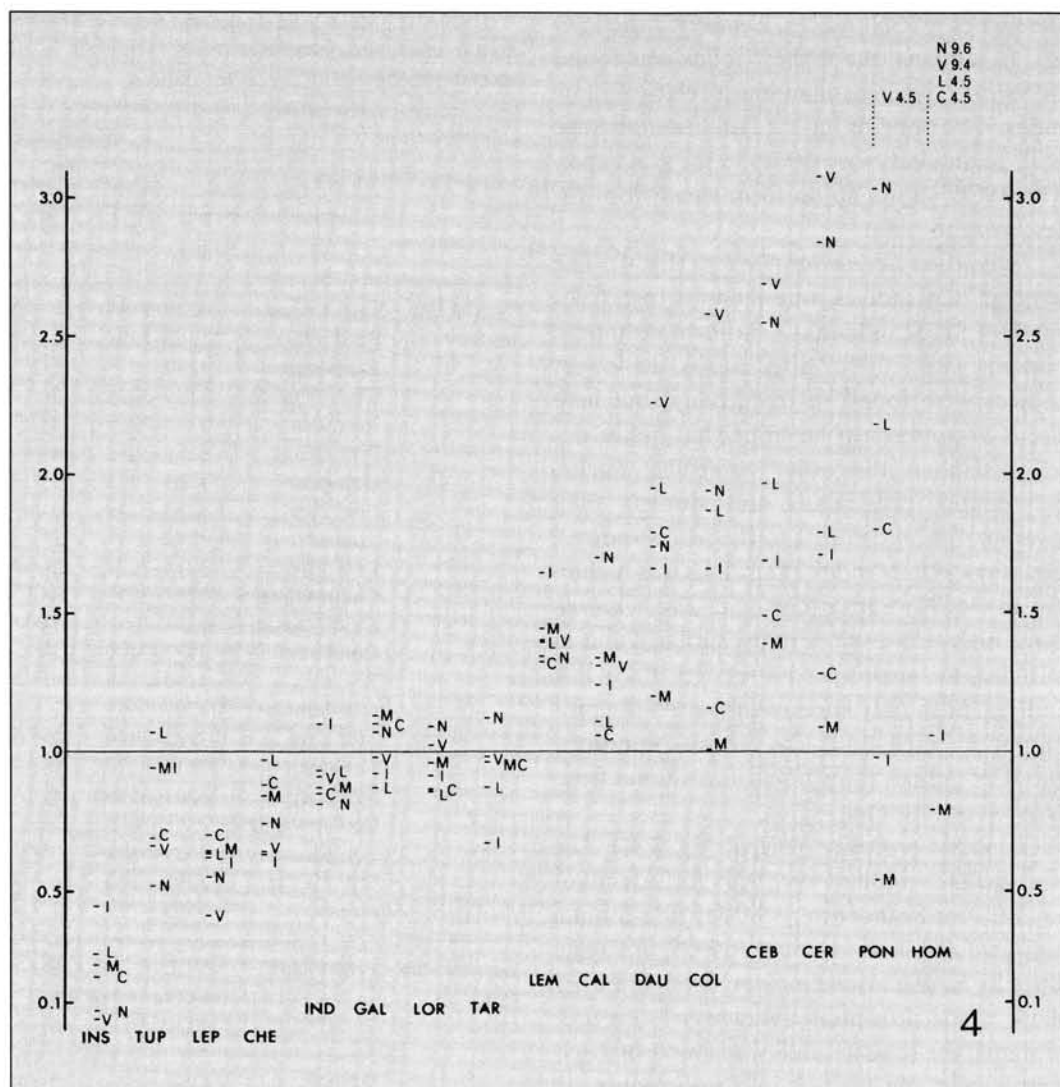


Fig. 2, 3. Indices of the medial (2) and lateral (3) cerebellar nuclei in taxonomic units. The indices are numerical expressions of the distances from the reference line (as e.g. given in figure 1 for LCN). Each of the small horizontal bars represents an index for a species. On the left, the range (vertical bars) and the mean (o) of the various taxonomic units are given, while on the right the sequences of the means are scaled.

Fig. 4. Means of neocortex (N), ventral pons (V), cerebellum (C), and of medial (M), interposed (I), and lateral (L) cerebellar nuclei. The taxonomic units are arranged from left to right according to ascending neocortex indices. CAL = Callitrichidae; CEB = Cebidae; CER = Cercopithecinae; CHE = Cheirogaleidae; COL = Colobinae; DAU = Daubentonidae; GAL = Galaginae; HOM = Hominidae; IND = Indriidae; INS = Insectivora; LEM = Lemurinae; LEP = Lepilemurinae; LOR = Lorisinae; PON = Pongidae; TAR = Tarsiidae; TUP = Tupaiidae.

the highest MCN indices, *Lepilemur* the lowest. In simians, the highest index was found in *Saimiri*, whereas man has a relatively low index. The pongids have a still lower average with particularly low values in the great apes. The MCN shows no uniform trend from Insectivora to man.

Interposed Cerebellar Nucleus (fig. 4). The average ICN indices were 0.44 for Insectivora, 0.94 for Scandentia, 1.53 for non-human simians, and 1.06 for man. Again, the lowest indices were found in Insectivora, but here the ICN appears to be somewhat better developed than the other cerebellar nuclei. Scandentia, prosimians and simians fully overlap, but in contrast to the MCN indices, the average ICN index of simians was higher. As in MCN, Lemurinae and *Daubentonia* have high ICN indices in prosimians; and in simians, the great apes have particularly low indices. Man only slightly surpassed the prosimian average, and, like the MCN indices, the ICN indices show no uniform trend from Insectivora to man.

Lateral Cerebellar Nucleus (fig. 3, 4). The average LCN indices were 0.27 for Insectivora, 1.07 for Scandentia, 1.80 for non-human simians, and 4.54 for man. Insectivora again have by far the lowest indices. Scandentia are close to the prosimian average (=1). In prosimians, the lowest value was found in *Lepilemur*, and the highest in *Daubentonia*. In simians, the distinctly lowest average was found for the Callitrichidae, and the clearly highest for man. There is a general trend for relative size of the LCN to increase from Insectivora to man, and the lateral nucleus is by far the most progressive of the three cerebellar nuclei. The similar trend found in the TCN is mainly due to the enlargement of the LCN, which in primates generally is the largest of the cerebellar nuclei (up to 2/3 in apes

Table III. Mean percentage and indices of taxonomic units and locomotor types, arranged in an ascending sequence of neocortex indices

			Species n
Taxonomic units	Insectivora ¹	0577-1137	10
	Scandentia	1263-1289	2
	<i>Lepilemur</i>	3239	1
	Cheirogaleidae	3199-3205	3
	Indridae	3243-3249	4
	Prosimians	3199-3282	18
	Galaginae	3265-3275	3
	Lorisinae	3253-3259	3
	<i>Tarsius</i>	3282	1
	Lemurinae	3215-3227	2
	Callitrichidae	3287-3311	4
	<i>Daubentonia</i>	3251	1
	Colobinae	3477-3499	3
	Primates	3199-3553	44
	NW Simians	3287-3379	12
	Monkeys	3287-3499	22
	Cebidae	3317-3379	8
	Simians	3287-3553	26
	Cercopithecinae	3391-3473	7
	Pongidae	3539-3551	3
	OW Simians	3391-3553	14
	<i>Homo</i>	3553	1
Locomotory types	Prosimians	VCL	9
		SCQ	3
		AQP	6
	Simians	OSB	3
		NAQ	9
		NSB	3
		OTQ	3
		OAQ	4
		OMB	2
		OTB	1
		SBP	1

Abbreviations as in table II.

Percentages				Indices						
TCN CER	MCN TCN	ICN TCN	LCN TCN	TCN	MCN	ICN	LCN	VPo	CER	NEO
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
2.81	23.8	40.9	35.3	0.31	0.23	0.44	0.27	0.04	0.19	0.07 ²
3.24	29.5	26.7	43.8	0.99	0.94	0.94	1.07	0.66	0.69	0.52
2.03	31.7	27.8	40.5	0.63	0.64	0.62	0.63	0.41	0.70	0.55
2.14	31.6	21.5	46.9	0.84	0.84	0.63	0.97	0.64	0.88	0.74
2.53	28.5	32.3	39.2	0.96	0.87	1.10	0.93	0.91	0.85	0.81
2.25	31.6	27.9	40.5	1.00	1.00	1.00	1.00	1.00	1.00	1.00
1.97	36.6	26.9	36.5	0.97	1.13	0.92	0.87	0.98	1.10	1.07
2.37	33.1	28.5	38.4	0.90	0.96	0.91	0.85	1.02	0.86	1.09
1.96	35.6	22.6	41.8	0.84	0.96	0.67	0.87	0.98	0.96	1.12
2.50	30.4	31.2	38.4	1.48	1.44	1.64	1.39	1.40	1.32	1.34
2.61	34.3	28.8	36.9	1.22	1.34	1.24	1.11	1.31	1.06	1.70
2.05	23.0	28.6	48.4	1.64	1.20	1.66	1.95	2.26	1.79	1.74
3.02	20.4	30.5	49.0	1.54	1.01	1.66	1.87	2.58	1.16	1.94
2.42	26.5	27.6	45.9	1.33	1.08	1.30	1.53	2.20	1.28	2.04
2.63	28.2	28.0	43.8	1.55	1.38	1.54	1.68	2.23	1.35	2.27
2.72	25.0	29.3	45.7	1.55	1.23	1.61	1.74	2.55	1.30	2.41
2.63	25.2	27.6	47.2	1.71	1.39	1.69	1.97	2.69	1.49	2.55
2.53	23.0	27.5	49.6	1.56	1.14	1.52	1.90	3.04	1.48	2.76
2.74	21.4	31.0	47.6	1.55	1.09	1.71	1.79	3.08	1.28	2.84
1.63	12.3	18.9	68.8	1.33	0.54	0.98	2.18	4.48	1.80	3.04
2.45	18.5	27.0	54.5	1.56	0.93	1.50	2.09	3.73	1.59	3.18
1.19	10.3	12.6	77.1	2.39	0.79	1.06	4.54	9.44	4.50	9.63
2.22	32.4	28.9	38.7	0.91	0.94	0.94	0.87	0.89	0.93	0.90
2.37	33.1	28.5	38.4	0.90	0.96	0.91	0.85	1.02	0.86	1.09
2.25	29.8	25.9	44.3	1.18	1.10	1.14	1.27	1.16	1.18	1.11
3.02	20.4	30.6	49.0	1.54	1.01	1.66	1.87	2.58	1.16	1.94
2.84	30.8	28.4	40.8	1.62	1.55	1.62	1.67	2.06	1.29	2.18
1.99	20.4	26.8	52.8	1.34	0.86	1.29	1.74	2.74	1.53	2.54
2.95	21.0	30.2	48.8	1.54	1.07	1.66	1.80	3.05	1.16	2.71
2.59	21.7	31.6	46.7	1.56	1.10	1.74	1.79	3.11	1.36	2.94
1.34	11.2	14.9	74.0	1.03	0.36	0.53	1.87	4.03	1.71	2.94
2.22	14.6	27.1	58.3	1.95	0.91	1.87	2.81	5.38	1.98	3.22
1.19	10.3	12.6	77.1	2.39	0.79	1.06	4.54	9.44	4.50	9.63

¹ In Insectivora the ventral pons is included in the cerebellum.

² The average index of the neocortex given for Insectivora in part I differs somewhat from the value given here due to different numbers of species (2 and 10, respectively).

and man), and thus considerably influences the total for the nuclei.

In conclusion, the lowest indices were always found in Insectivora, and the highest in simians, namely in *Saimiri* (MCN, ICN) and *Homo* (LCN). Prosimians fully overlapped with simians in MCN and ICN and partly in LCN. A simian average close to 1 was found in MCN, whereas simians had higher indices in ICN (1.5) and particularly in LCN (1.9). In the LCN, man distinctly surpassed the range of variation for non-human simians. In contrast, man's index for the MCN was distinctly lower than in prosimians and simians on the average. In the ICN, man's index was close to the prosimian average.

Interstructural Comparisons

On the basis of interstructural comparisons (percentages, interstructural regression), it is hard to decide whether the resulting differences are due to the structure itself or to the reference system. Percentages are given in table II, and averaged in table III (columns 1–4); correlation coefficients and slopes of interstructural regression lines based on the overall material are summarized in table IV. High coefficients may indicate close functional interrelationships, but may also be due to similar evolutionary trends. The slopes show which of the two plotted structures has the greater increase and how large the difference is.

Cerebellar Nuclei Relative to Cerebellum (TCN/CER). Expressed as a percentage of the cerebellum, the TCN ranges from 1.16% in *Gorilla gorilla* to 3.95% in *Callicebus*. The average percentages are 2.81 for Insectivora, 3.24 for Scandentia, 2.25 for prosimians, 2.59 for non-human simians and 1.19 for man. There is some overlap between Scandentia (high values), Insectivora (medium)

and prosimians (low), whereas simians cover the whole range. The percentage of man was found to be the lowest next to *Gorilla*. There is no uniform trend from Insectivora to man.

In the log TCN versus log CER regression line, the slope is close to 1 (table IV), thus indicating that with enlarging cerebellum the cerebellar nuclei enlarge in a similar proportion. However, it is noteworthy that in simians the three lowest percentages are in the great apes and in man. In these forms the cerebellar nuclei occupy a relatively small portion of the cerebellum.

Medial Cerebellar Nucleus Relative to Cerebellar Nuclei (MCN/TCN). The MCN expressed as a percentage of the TCN varies from 9.7% in the chimpanzee to 37.5% in *Galagoides*. There is a wide and, in places, complete overlap between the taxonomic groups. The highest averages were found in prosimians and Scandentia. In prosimians, high values were found in Galaginae, *Loris* and *Tarsius*; a very low value was found in *Daubentonia*. In simians, the clearly highest average was found in Callitrichidae; high values exist also in some Cebidae (*Callicebus*, *Saimiri*, *Aotus*). The lowest percentages were found in apes and man, and there was no overlap with other species.

In the log MCN versus log TCN regression line, the slope is 0.923 (table IV), indicating that with enlarging cerebellar nuclei the MCN has a lower enlargement than the total of the cerebellar nuclei. Deviations of Insectivora at the lower end of the line and of the great apes and man at the upper end are in the same direction, suggesting a relatively small MCN in the complex of the cerebellar nuclei.

Interposed Cerebellar Nucleus Relative to Cerebellar Nuclei (ICN/TCN). The ICN expressed as a percentage of the TCN ranges

from 12.6% in man to 50.6% in *Echinops*. Within Insectivora, the larger spiny forms (Tenrecinae, *Erinaceus*) have an average of 45.8%, and the smaller hairy forms (Soricidae) an average of 36.0%. No overlap between these two groups was found. Of the Insectivora, only *Sorex araneus* reached into the primate range. Scandentia, prosimians, and non-human simians fully overlap and their averages are similar. In prosimians, the highest percentages were found in Indridae and Lemurinae, the lowest in Cheirogaleidae and in *Tarsius*. In marmosets, monkeys and gibbons, the percentages are relatively uniform, ranging from 22 to 34%, whereas the values of the two great apes and man are about half of the middle of this range. Man does not overlap with other groups.

In the log ICN versus log TCN regression line, the slope is 0.934, and similar to that of log MCN versus log TCN; this also suggests a lower enlargement of the ICN relative to the total of the cerebellar nuclei.

Lateral Cerebellar Nucleus Relative to Cerebellar Nuclei (LCN/TCN). The LCN expressed as a percentage of the TCN varies from 26.7% in *Echinops* to 77.1% in man. The ranges of the taxonomic groups overlap widely. In Insectivora, the spiny and hairy groups (see ICN) differ also in the percentage of the LCN without overlap. The averages are 31.1 and 39.4%, respectively. In prosimians, high percentages were found in Cheirogaleidae and *Daubentonia*, low percentages in Galaginae, and an especially low percentage in *Loris* (member of the Lorisinae). In simians, the lowest percentages were found in Callitrichidae and some Cebidae (*Callicebus*, *Saimiri*), the highest in the apes and in man.

In the log LCN versus log TCN regression line, the slope is 1.074, indicating a higher enlargement of the LCN relative to the total

Table IV. Slopes of the interstructural regression lines and correlation coefficients (r^2) (in italics) for the overall material ($n = 56$, i.e. 10 species of Insectivora, 2 of Scandentia, 18 of prosimians, 26 of simians)

X Y	CER ¹	VPo ²	TCN	MCN	ICN
LCN	1.031 <i>0.990</i>	0.795 <i>0.971</i>	1.074 <i>0.994</i>	1.127 <i>0.955</i>	1.125 <i>0.966</i>
ICN	0.884 <i>0.953</i>	0.659 <i>0.906</i>	0.934 <i>0.985</i>	0.989 <i>0.963</i>	
MCN	0.876 <i>0.952</i>	0.593 <i>0.938</i>	0.923 <i>0.976</i>		
TCN	0.955 <i>0.986</i>	0.711 <i>0.964</i>			
VPo ²	1.290 <i>0.969</i>				

¹ In Insectivora the ventral pons is included in the cerebellum.

² VPo measurements in 2 species of Insectivora only.

of the cerebellar nuclei. When regressing LCN versus MCN or ICN the slopes are of course still higher (1.127 and 1.125; see table IV).

Discussion

The various species and taxonomic groups differ in their behavioral patterns; ecoethological groups may correspond to taxonomic units or go across them. In previous investigations [e.g. Stephan et al., 1982, 1984, 1985; Baron et al., 1983; Frahm et al., 1984], differences in the relative size of structures could be related to locomotor patterns, preferred diets and activity timing. Therefore, attention will be paid here to such behavioral properties. The affiliation of the species to

the various ecoethological types is given in table I.

Data on ventral pons (VPo), cerebellum and neocortex, taken from part I, will be included; in general, average values will be used.

Locomotor Types within Various Taxonomic Groups

The *Insectivora*, which are conservative quadrupedal forms, have the lowest relative size in all the structures under comparison (table III). The lowest indices were found for the VPo and neocortex. All indices are clearly below the prosimian average (=1), and even the highest (0.44 for the ICN) is less than 1/2 of that average. Surprisingly, the index of the MCN is low and similar to that of the LCN. The larger spiny forms (Tenrecinae, *Erinaceus*) have relatively larger ICN and smaller LCN than the smaller hairy forms.

The VPo index is closer to that of the neocortex than to that of the cerebellum and is consistent with the general trends in primates. This results also from the VPo:neocortex ratio (part I), which is similar to that of primates. In contrast to the similar low relative size of VPo and neocortex, the cerebellum index is about 4 times higher (0.19 vs. 0.04 and 0.05; see part I, table II), and in this respect the *Insectivora* differ considerably from the primates. Because of the relatively large cerebellum, the VPo:cerebellum ratio is the lowest among all the taxonomic groups. The brain stem appears to be relatively large in *Insectivora*, since the percentage of VPo relative to this structure is by far the lowest found in all our material: less than 1/200 of the brain stem is VPo. All data point to a primitive status of locomotor ability among *Insectivora* in comparison to the other groups under investigation.

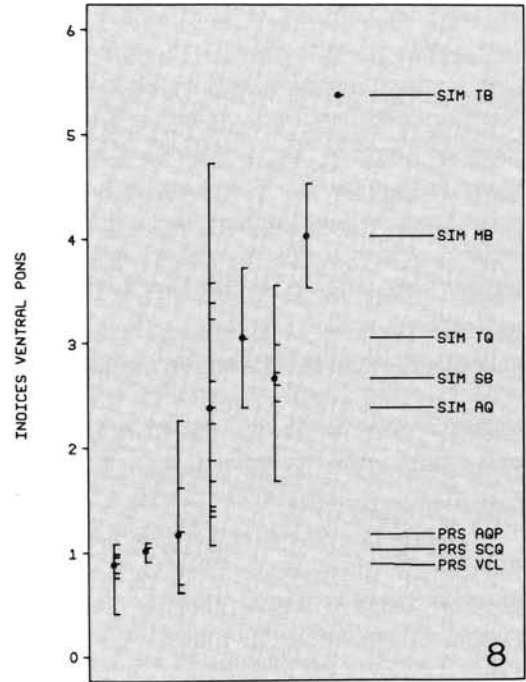
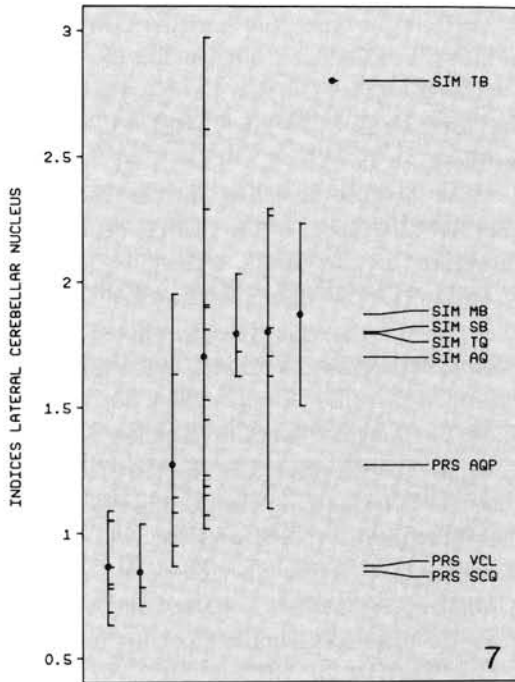
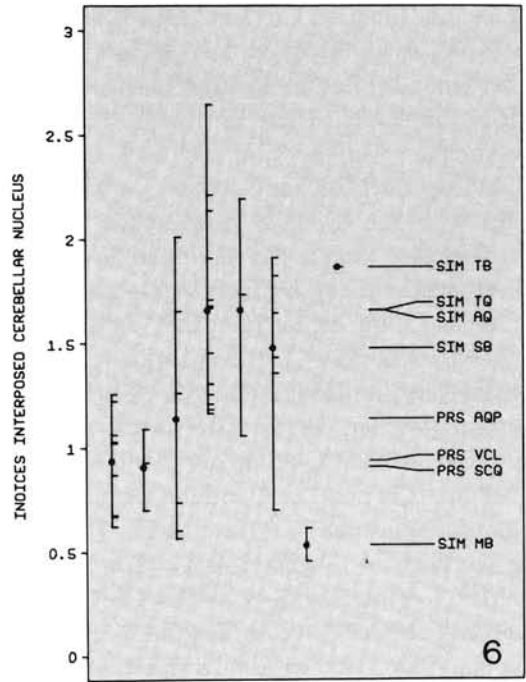
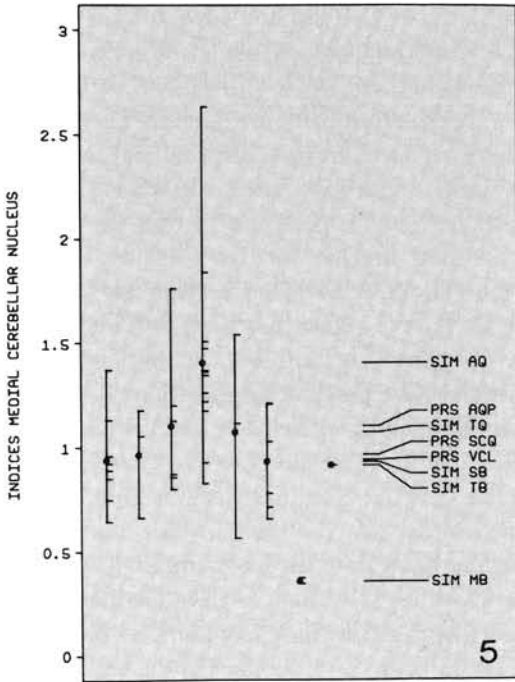
Scandentia (tree shrews) are the only one of all the taxonomic groups under investigation with distinctly higher indices for all the cerebellar nuclei, as compared to those of cerebellum, VPo and neocortex. The LCN index is slightly above the prosimian average (=1). VPo and cerebellum indices are about 2/3 of the prosimian average; both indices are distinctly larger than that of the neocortex. This is reflected also in the high VPo:neocortex ratio, which is one of the highest in all the taxonomic groups (see part I, table II).

Tree shrews are quickly moving terrestrial and arboreal quadrupeds. They use their hands, which are convergent but not prehensile, when manipulating food [Napier and Napier, 1967]. Thus they differ from *Insectivora* and Primates in their locomotor abilities.

The *prosimian primates* may be classified in three main categories of locomotion: vertical clingers and leapers (VCL) [Napier and Walker, 1967], comprising *Lepilemurinae*, *Indriidae*, *Galaginae* and *Tarsius*; arboreal quadrupeds (AQP), comprising *Cheirogaleidae*, *Lemurinae* and *Daubentoninae*; and slow-climbing quadrupeds (SCQ), comprising *Lorisinae* [see Napier and Napier, 1967].

There are no large differences in the average indices among the three main groups of locomotor patterns (fig. 5–8). Vertical clin-

Fig. 5–8. Indices of the medial (5), interposed (6), and lateral (7) cerebellar nuclei and of the ventral pons (8) in various locomotor types of primates. PRS AQP = Prosimian arboreal quadrupeds (n = 6); PRS SCQ = prosimian slow-climbing quadrupeds (n = 3); PRS VCL = prosimian vertical clingers and leapers (n = 9); SIM AQ = simian arboreal quadrupeds (OAQs + NAQs; n = 13); SIM MB = simian modified brachiators (n = 2); SIM SB = simian semibrachiators (OSBs and NSBs; n = 6); SIM TB = simian true brachiator (n = 1); SIM TQ = simian terrestrial quadrupeds (n = 3).



gers and leapers (VCL) have the lowest average indices in neocortex, VPo and MCN, and the second (middle) position in cerebellum, ICN and LCN. There is always, however, complete overlap with the slow-climbing quadrupeds (SCQ), and generally wide overlap with the arboreal quadrupeds (AQP). *Galagoides*, which has the best developed cerebellar nuclear complex and the highest indices for the cerebellum, the VPo and the neocortex of all VCLs, also seems to be the most versatile locomotor type of the whole group [Walker, 1979]. The lowest average indices in the SCQ group are found for cerebellum, ICN and LCN, and the second (middle) for neocortex, VPo and MCN. The VPo index is distinctly higher than the cerebellum index. It is interesting to note that SCQs have a relatively well-developed striatum [Stephan et al., 1985]. This suggests a greater involvement of the higher centers in the motor activity of the Lorisinae. The stealthy, slow-climbing locomotion in the upper layers of the canopy by *Perodicticus* is made possible only by very finely tuned movements [see Walker, 1969]. Similar motor capacities have been reported for *Nycticebus* by several authors [Ishida et al., 1983; Wrobel, 1966]. AQPs in all cases have the highest average indices. This, however, is mainly due to two single species *Daubentonia*, *Lemur*, which generally have very high indices. If these two species are excluded, those remaining correspond in range and average with the VCLs and SCQs. It is only in the LCN that the remaining AQP species maintain a slightly higher average index.

In all *simians*, the average VPo indices are higher than those of the cerebellum and are generally close to those of the neocortex. The indices of the cerebellar nuclei vary mostly around those of the cerebellum. On the aver-

age, the MCN has the lowest indices, the LCN the highest, and the ICN intermediate values (fig. 4).

The main locomotor patterns of *non-human simians* are quadrupedalism and brachiation, with subdivisions in terrestrial and arboreal quadrupeds and in true, semi- and modified brachiators (fig. 5–8) [see Napier and Napier, 1967]. The quadrupeds ($n = 16$; table I) have higher indices in MCN and ICN (with partial overlap) than the other groups, whereas the brachiating species ($n = 9$) have slightly higher indices in cerebellum, VPo and LCN. There is, however, always complete overlap. No difference was found in the neocortex. In the MCN, the brachiating forms have an even lower index (0.8) than the average prosimian ($= 1$). In the quadrupedal forms, all the indices under consideration except for MCN are higher in the Old World than in the New World monkeys. There are, however, several exceptions. *Cebus*, which is an extremely agile and versatile animal exhibiting considerable manipulative skill in handling objects [Napier and Napier, 1967], has the highest LCN index of the non-human primates.

Within the brachiating species, there are marked differences in the relative size of the structures in the semi-, true and modified brachiators. The semibrachiators ($n = 6$; table I) have the lowest average indices for neocortex, cerebellum, VPo and LCN, the highest for MCN and an intermediate position for ICN. The range of variation is generally wide, since in most structures there are marked differences between and within the New World semibrachiators (NSB) and the Old World forms (OSB) (three species each). In the OSBs, the indices of neocortex and cerebellum are markedly smaller than in the NSB group, whereas the MCN and ICN are larger,

and the VPo and LCN similar. The true brachiator (OTB) *Hylobates* has by far better developed cerebellar nuclei and VPo than the modified brachiators (OMB) *Pan* and *Gorilla*. The effortless ease of gibbons in 'flying' through the trees is in accordance with the high indices of the progressive motor structures. In neocortex, VPo and LCN, the chimpanzee has markedly higher indices than the gorilla, whereas in cerebellum, MCN and ICN, the differences are small. The relatively smaller indices in the gorilla correspond certainly to its lesser versatility, but may also be influenced by its exceeding body weight [see Frahm et al., 1982].

The locomotor patterns of *hominids* may be characterized as specialized bipedal (SBP). In man, the indices of LCN, cerebellum, VPo, and neocortex are by far the highest, and the gap between the indices of man and those of non-human primates is in general very large. For MCN and ICN, the indices were below the average values found in simians, and for the MCN, also below that of prosimians. In contrast, the LCN in *Homo* was by far the relatively largest among all primates. The high LCN index is similar to that of the cerebellum, whereas the indices of VPo and neocortex are more than twice as large. Because of the extremely large neocortex, the VPo:neocortex ratio is slightly below the average of simians and primates as a whole. The VPo:cerebellum ratio is close to the simian average, but the VPo in percentage of the brain stem is extremely high. In man, more than 1/3 of the brain stem is VPo. Recent man is not able to 'fly' like gibbons through trees, but its erect posture and extremely versatile and coordinated finger movements are far superior to the possibilities of other primates, thus explaining the extremely high indices in neocortex, VPo, cerebellum, and LCN.

Dietary Habits

A classification of the species according to their foraging techniques was given by Baron et al. [1983]. Based on investigations by Hladik [1975, 1979], Baron et al. [1983] distinguished three grades: Insectivorous forms which supplement their diet with various amounts of fruits and gum (grade 1); frugivorous species which include the preferential gum- and seed-eaters that eat animal or vegetable diet as a protein supplement (grade 2); and folivores obtaining their protein mainly from leaves, shoots or flowers, with a supplement of various amounts of fruit (grade 3).

With regard to the average indices of the various dietary groups, the mainly vegetarian species of prosimians generally have lower indices, and the insectivorous and carnivorous species higher values. However, the predominantly folivorous *Lemur* and *Varecia*, and the frugivorous *Galago* deviate from this general rule with high indices. The relatively high indices of *Galago* may be the consequence of the food sources, which are widely dispersed or have a patchy distribution and force the animals to travel over considerable distances in a complex environment. *Galago senegalensis* covers a distance of about 2 km every night [Charles-Dominique and Bearer, 1979]. Among insect-eating prosimians, almost every species is a distinctive ecoethological type. Similar food preference seems to have little influence on the relative size of the motor structures, and other ecoethological influences, e.g. locomotor adaptation, seem to have a greater effect.

In simians, the folivorous species have lower average indices than the frugivorous species. The more dispersed distribution of the food sources in the frugivorous species and the greater ballast in the intestines of the

folivorous species may account for these differences. Complete overlap in the range of variation and only slight differences are found in VPo, ICN and LCN, whereas in the neocortex, cerebellum, and MCN, the differences are more distinct.

Activity Timing

With regard to the average indices in relation to differences in the periods of main activities, the diurnal prosimians have higher average values than the nocturnal prosimians. Nocturnal species may have more restrictions to mobility than diurnal forms. However, full overlap and only slight differences were found in neocortex, cerebellum, VPo, and MCN, whereas in ICN and LCN the differences are somewhat greater. In the simians, only one species (*Aotus*) is nocturnal, and compared with the average indices of the diurnal species, its indices are lower in all structures except MCN. However, *Aotus* is always within the range of variation of the diurnal simians.

Trends in Size Differences of Motor Structures and Functional Significance

Medial Cerebellar Nucleus (fig. 5). In primates, the MCN has a larger relative size in the quadrupedal than in other groups. This is true both in prosimians and simians, but there is always a wide overlap. In simians, the brachiating forms tend to have relatively smaller MCN. The modified brachiators and specialized bipedals (great apes, man) in particular have relatively very small indices. Furthermore, in the terrestrial quadrupeds there is a slight tendency for the MCN to be somewhat smaller than in the arboreal quadrupeds, and in the Old World arboreal quadrupeds to be smaller than in the New World forms.

The MCN may reflect the relative size of the medial zone, and thus may give some indications as to the relative size of the cerebellar vermis. Macromorphological comparison of the brain of Old World and New World arboreal quadrupeds and the low relative size of the vermis in man (see part I) support this view.

The fact that the medial zone seems to be mainly concerned with regulation of major changes in postural tone, equilibrium, and locomotion of the entire body including the limbs [Chambers and Sprague, 1955a] is consistent with our findings that quadrupeds tend to have a larger MCN than the brachiating forms and that within quadrupeds, species of the branch-running arboreal type have somewhat better developed MCN than species of the ground-running terrestrial type.

Interposed Cerebellar Nucleus (fig. 6). Both in prosimians and simians, the quadrupedal forms again have slightly larger ICNs; there is, however, complete overlap in prosimians and a wide overlap in simians. No differences were found between arboreal and terrestrial quadrupeds. The modified brachiators (great apes) again have very low indices, that of man also is low, while that of the true brachiator (*Hylobates*) is relatively high.

The ICN may reflect the relative size of the intermediate zone. Since there are no macromorphological indications for this zone and no structural differences, which would allow volume measurements, the relative size of the ICN is so far the only indication of the relative size of the intermediate zone.

Experimental studies have revealed that the paravermal-interpositus zone plays a role in controlling movements and postural reflexes of the ipsilateral limbs as well as of the

shoulder and neck [Rispa-Padel et al., 1982]. It particularly controls postural placing and hopping reflexes [Chambers and Sprague, 1955b]. These findings may explain the slight increase of the ICN in quadrupedal primates and especially in the brachiating *Hylobates*, since chiefly the forelimb extension seems to be evoked by paravermal stimulation [Goodman and Simpson, 1961].

Lateral Cerebellar Nucleus (fig. 7). No differences in the relative size of the LCN between quadrupedal and brachiating forms of simians were found, whereas in prosimians, the arboreal quadrupeds tended to have slightly higher indices than the other groups.

The LCN may reflect the size of the lateral zone and thus may give some indication as to the size of the cerebellar hemispheres. Macromorphological comparison of the hemispheres in prosimians, monkeys, apes and man reinforce the impression that a strong increase has occurred. The enormous increase of the lateral or hemispherical zone correlates well with the acquisition, in primates, of planned, skilled movements, especially of finely tuned hand and finger movements, during evolution. In simians, the *Saguinus* species have the lowest LCN indices, and their hands, with a non-opposable thumb, have only a limited range of manipulative function [Napier and Napier, 1967]. Species such as *Cebus*, *Hylobates* and *Pan*, however, which are known to be capable of very versatile manipulative activities, have very high LCN indices, and man has an extremely high index. Comparative studies on the development of cerebellar nuclei conducted by Ogawa [1935] in Pinnipedia and Cetacea, by Brauer [1966] in Canioidea, and by Baron [1978] in Chiroptera have shown that the relative size of the LCN corresponds to the functional importance of the limbs.

Cerebellum. In prosimians, the arboreal quadrupeds have a slightly larger cerebellum than other groups. In the lorises, the slowly-moving *Lorisinae* and *Otolemur* have distinctly lower cerebellum indices than the fast-moving *Galagoides* and *Galago*. In simians, the cerebellum is slightly larger in brachiating than in quadrupedal forms. There is, however, complete overlap. In simians, a relatively large cerebellum is found in true brachiators and modified brachiators, as well as in the frugivorous New World semibrachiators (*Ateles*, *Lagothrix*). However, compared with the non-human primates, man has by far the largest cerebellum.

In Chiroptera the relative size of the cerebellum was found to be related to the width of the spectrum of motor abilities rather than to narrowly specialized motor patterns, even though they permit nimble movements [Stephan and Pirlot, 1970; Stephan et al., 1974]. This could also be inferred from Portmann's [1947] work on birds, and is obviously true for primates as well, as shown by the modified brachiators and especially by man, who exhibits the most skilled and diversified movements.

Ventral Pons (fig. 8). The VPo indices are closer to those of the neocortex than to those of the cerebellum. Certainly the strong enlargement of the VPo is caused in addition to increasing cortico-pontine fibers by the increasing amount of passing cortico-spinal fibers (pyramidal tract). Both contribute to the strong parallelism between VPo and neocortex. In simians, the VPo is on the average slightly larger in brachiating than in quadrupedal forms, caused mainly by the high indices in true and modified brachiators. Man has by far the highest index followed by the true brachiator. Low values are found in the New World arboreal quadrupeds and in the

folivorous NSB *Alouatta*. Like the LCN, the VPo reflects the size of the cerebellar hemispheres and thus the lateral zone. Although pontocerebellar fibers terminate in all parts of the cerebellar cortex, except perhaps the flocculonodular lobe, the neocerebellum (mainly lateral zone) receives most of the pontocerebellar fibers [Brodal and Jansen, 1946].

Neocortex. When the average indices of the various motor patterns within prosimians were compared, no differences were found. In simians too, quadrupedal and brachiating forms generally have similar neocortex indices with a complete overlapping range of variation. Within the brachiating simians, true and modified brachiators have slightly larger averages than semibrachiators, and within the quadrupedal simians, terrestrial, slightly larger than arboreal. The folivorous semibrachiators (Colobinae, *Alouatta*) have distinctly lower indices than the frugivorous forms (*Ateles*, *Lagothrix*). Man has by far the highest index.

Evolutionary Trends

As stated in the Discussion of part I, various parts of the cerebellum developed at different rates from Insectivora or prosimians to man. Macromorphological observations suggest that the cerebellar hemispheres enlarged much more than the non-hemispheric parts. This impression was reinforced by quantitative and mathematical approaches, in which the cerebellar hemispheres relative to the whole cerebellum increase from about 41% in the prosimian *Microcebus* to 88–89% in man. Consequently, the percentage of the non-hemispheric parts of the cerebellum, which includes mainly the vermis, decrease

from about 59% in *Microcebus* to 11–12% in man. For man's cerebellar hemispheres, an index of about 10 was calculated, and for the non-hemispheric parts, the value was 0.9 (prosimian base).

The brain components measured that are involved in motor activities may be grouped according to their trends and intensities and with special reference to their indices in man as follows: (1) high progression in neocortex and VPo with indices around 9.5, (2) medium progression in cerebellum and LCN with indices around 4.5, and (3) low progression or even slight regression (i.e. indices around 1) in ICN and MCN. From the calculations above, the cerebellar hemispheres correspond to the first group, the non-hemispheric parts to the third.

As mentioned in the Introduction, the cerebellum and its nuclei may be subdivided from a functional point of view into three longitudinal zones: the vermis-medialis, the pars intermedialis-interpositus, and the hemisphere-lateralis component. VPo is a relay station between the neocortex and the cerebellum, mainly the cerebellar hemispheres [Brodal and Jansen, 1946; Ito, 1984], and is thus primarily related to the lateral zone. From the evolutionary trends described above it is likely that the lateral zone is the most progressive and the medial zone the least progressive. The intermediate zone may have an intermediate progression. However, there is no uniformity of the indices within the most progressive lateral zone, since the indices of the LCN tend to be lower than those of the VPo and cerebellar hemispheres, and also those of the neocortex. Obviously, the structures embodied in the afferent limb mediating information flow to the cerebellar integrator, increase more than the structures on the output side. Differences between the

input and output sides, which could be similarly interpreted, were described by Zilles et al. [1982] for the area striata.

The close functional connections between the structures of the afferent limb of the lateral zone are paralleled by similar size indices and high interstructural correlations. Such interdependence was also shown, for example in structures within the limbic system [Stephan, 1975]. However, high interstructural correlations may also exist when the functional connections are less strong, e.g. between VPo on the one hand and medulla oblongata and mesencephalon on the other (part I, table III). In such cases, they may be the consequence of similar evolutionary trends. The evidence of correlation coefficients was discussed in more detail by Jolicoeur and Baron [1980] and Jolicoeur et al. [1984] on the basis of volume measurements of brain structures in Chiroptera, Insectivora and Primates.

General Conclusions

The evolution of the cerebellar complex in primate phylogeny is characterized by a strong enlargement of the components of the lateral zone, especially of the afferent limb of the cerebellar cortex, whereas those of the medial zone are much less progressive and seem to be even relatively larger in prosimians and monkeys than in apes and man (fig. 4). It is reasonable to suppose that the overall trends reflect a general progression in motor behavior from simple movements to highly complex and organized movements, especially of the hands and fingers. Differences in the relative size of the motor structures due to differences in ecoethological characters, such as motor patterns, feeding strategies and activity timing, can hardly be separated from the general trends. In the case of different ecoethological adaptations at a similar evolutionary level, differences in the

relative size of the structures are absent or small, and in the case of different evolutionary levels but similar ecoethological characters, e.g. the same type of food or similar activity timing, there are large differences in the relative size of the motor structures. Obviously, feeding strategies and foraging techniques differ between prosimians and simians despite the fact that, for example, the same type of food is eaten. Prosimians execute hand movements of a fairly fixed pattern which limits the variability of food manipulation. Simians execute much more versatile finger movements [Baron et al., 1983]. The importance of sharply tuned voluntary movements increases in higher evolutionary levels and obviously is a substantial component of general evolution. This is confirmed by the very high position of man.

Our results confirm the morpho-functional and physiological findings, which indicate a functional diversity of the three cerebellar nuclei and the corresponding zones of the cerebellar cortex. The vermis-medialis component is concerned with postural tone, equilibrium, and locomotion of the entire body. The intermediate-interpositus zone is considered to have more discrete control of skilled movements of the proximal limb musculature, whereas the hemisphere-lateralis component is concerned with skilled movements of the distal parts of the limbs [see Woodburne, 1967]. The lateral zone is the most progressive and the medial zone the least progressive. Our data do not allow, however, deductions about trends in the second fundamental component of the cerebellar complex, which is the flocculonodular lobe (vestibulocerebellum). Its comparative size may be deduced from measurements of the vestibular nuclei. Such investigations were presented in relation to flight behavior

in Chiroptera by Baron [1977] and are now in preparation for Primates. Another quantitative study on the vestibulocerebellar complex was presented by Matano et al. [1982]. These authors compared the size of the semicircular canals in *Tarsius bancanus* and *Nycticebus coucang* in relation to leaping behavior.

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